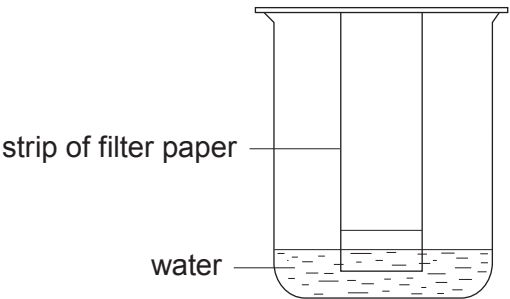
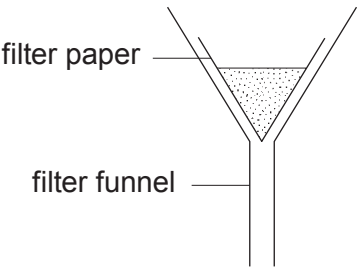


Answer all questions.

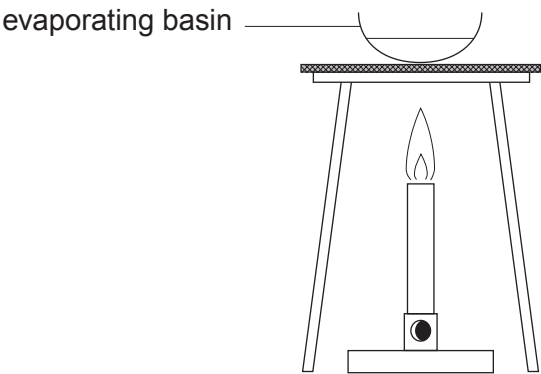
1. (a) The diagrams show three methods, **A**, **B** and **C**, used to separate mixtures.



A



B



C

Give the letter, **A**, **B** or **C**, of the method you would use to:

(i) remove sand from sea-water,

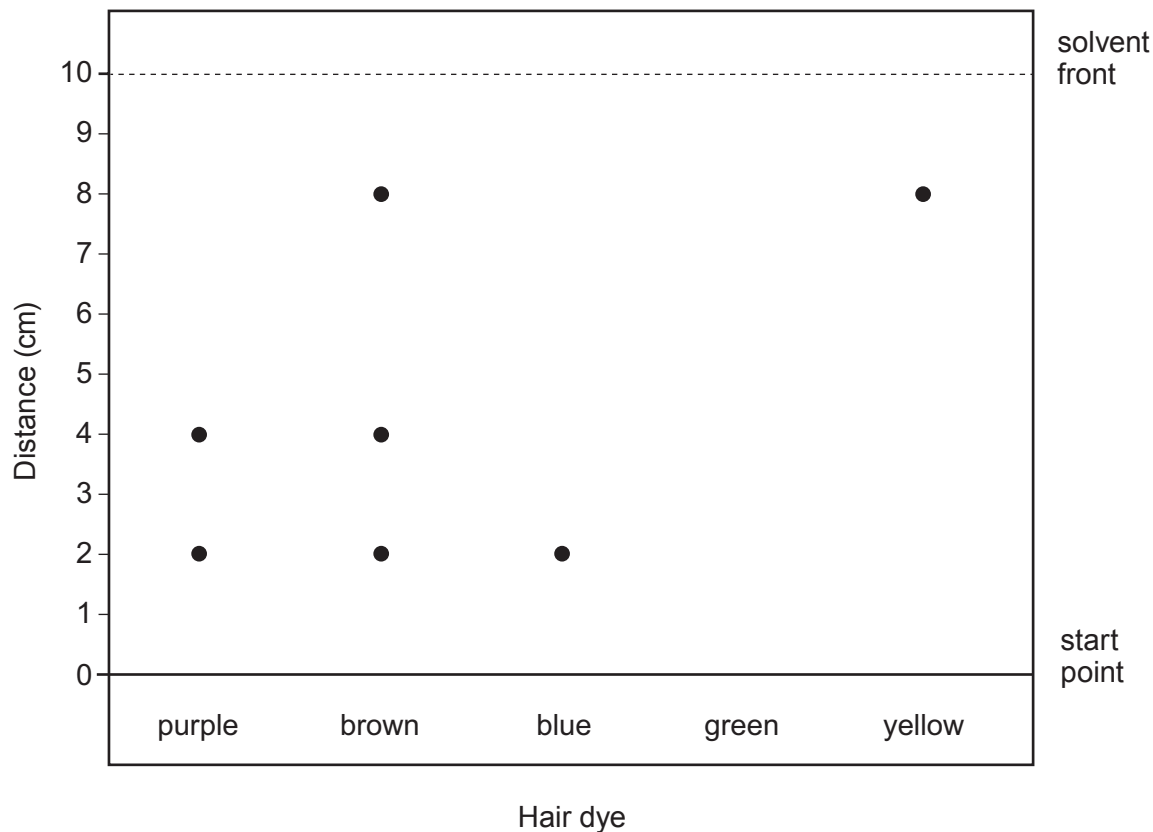
[1]

(ii) show that sea-water contains salt.

[1]



(b) Paper chromatography was used to show the pigments present in different hair dyes.



(i) Give the **two** dyes that are mixed to make brown hair dye. [1]

..... and

(ii) Green hair dye is made by mixing together blue and yellow dyes.
Draw on the diagram the result that you would expect to see for green hair dye. [1]

(iii) The R_f value of a substance can be used to identify that substance.
The R_f value of the pigment in red hair dye is 0.4.
Use the equation below to calculate the distance this red hair dye would have moved on the diagram. [2]

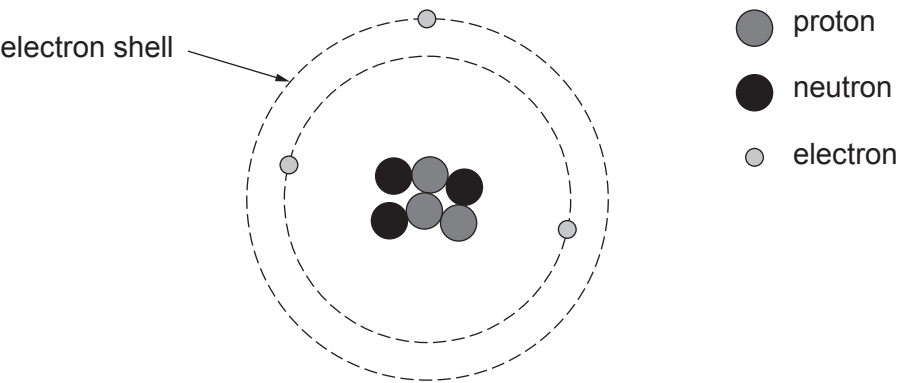
distance moved = $R_f \times$ distance moved by solvent

distance moved = cm



Answer all questions.

1. (a) The diagram represents a lithium atom. Its nucleus contains protons and neutrons.



(i) Complete the table of information about particles in the atom. [2]

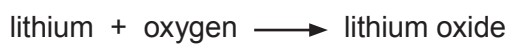
Name of particle	Charge	Mass
proton		1
neutron	0	
electron	-1	almost zero

(ii) Use the information in the diagram to give the mass number of this lithium atom. [1]

.....



(b) Lithium reacts with oxygen. The word equation for this reaction is as follows.



(i) Name the **metal** in this equation. [1]

.....

(ii) Name the **compound** in this equation. [1]

.....

(iii) Name a **reactant** in this equation. [1]

.....

(iv) Lithium oxide contains the ions Li^+ and O^{2-} .

Underline the formula of lithium oxide. [1]



1. Elements **A**, **B**, **C**, **D** and **E** are shown in the Periodic Table.

[illegible]

[5]

	Letter
is in Group 2	
makes a 'pop' noise with a lit splint	
has both metal and non-metal properties	
reacts with lithium to make lithium chloride	
is unreactive	

Answer **all** questions.

1. (a) Rock salt is a mixture of salt and sand. Crystals of pure salt can be obtained from rock salt.

A–E are the steps in the method used but they are in the wrong order.

- A** Add water to a sample of rock salt in a beaker and stir
- B** Heat the solution to evaporate some of the water
- C** Grind the rock salt into a fine powder
- D** Filter the mixture to separate the sand from the salt solution
- E** Leave the saturated solution in a warm place for a few days so that crystals of salt form

Put the steps in the correct order. The first step is already included.

[2]

C

first step

- (b) A student was asked to investigate the dyes present in an orange sweet.

The student carried out the following method. There are two errors in the method.

- Draw a line using a ruler and pen on chromatography paper.
- Place a sample of the orange colour on the line.
- Stand the chromatography paper in a beaker and add enough water to just cover the sample.
- Leave the paper to stand until the water rises to the top of the paper.

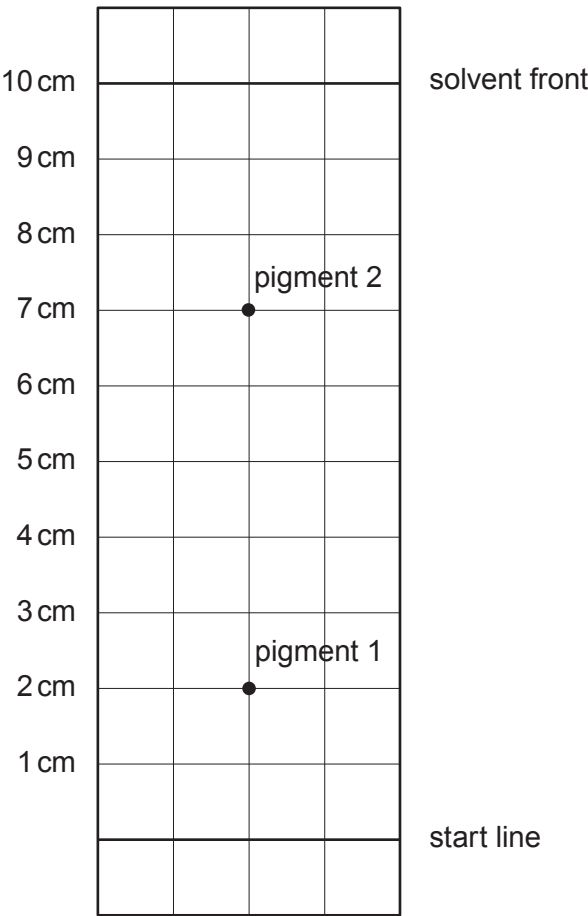
- (i) State the **two** errors in the method.

[2]

1.
2.



(ii) Another student used a correct method and obtained the chromatogram below.



Use the formula to calculate the R_f value for pigment 2. [2]

$$R_f = \frac{\text{distance travelled by the pigment}}{\text{distance travelled by the solvent front}}$$

$R_f =$

6



Answer **all** questions.

1. (a) The table shows four compounds and their formulae.

Compound	Formula
nitrogen dioxide	NO_2
hydrogen bromide	HBr
ethane	C_2H_6
ammonia	NH_3

Use the information in the table to answer the following questions.

- (i) Give the
- name**
- of the compound that contains the elements hydrogen and bromine. [1]

.....

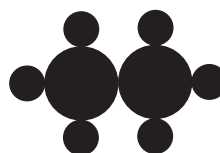
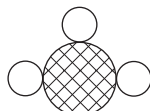
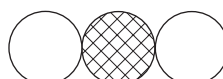
- (ii) Give the
- names**
- of the elements present in ethane. [1]

..... and

- (iii) Give the
- total**
- number of atoms found in a molecule of nitrogen dioxide. [1]

.....

- (iv) Tick (✓) the box next to the diagram that represents a molecule of ammonia,
- NH_3
- . [1]


☐

☐

☐

☐


- (b) Some ants spray a toxic mixture containing formic acid to defend themselves from predators.

Each molecule of formic acid contains two atoms of hydrogen, one atom of carbon and two atoms of oxygen.

Use this information to give the formula of formic acid.

[1]

.....

- (c) The box contains the names of three compounds that are made from positive and negative ions.

lithium chloride

magnesium bromide

calcium oxide

- (i) Give the formulae of the **ions** present in lithium chloride.

[1]

..... and

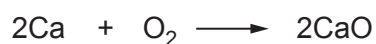
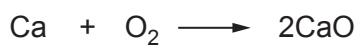
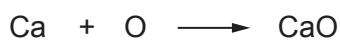
- (ii) Give the formula of magnesium bromide.

[1]

.....

- (iii) Tick (✓) the box next to the correct equation for the reaction between calcium and oxygen to produce calcium oxide.

[1]


☐

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